

JOEL NYONGESA

Data Scientist | Data Analyst

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PROFILE

I am a data scientist with a strong foundation in mathematics, statistics, and machine learning, currently pursuing an MSc in Data Science at AIMS Senegal (Mastercard Foundation Scholar). I build machine learning pipelines and analyse complex datasets from air quality forecasting to agricultural market intelligence to generate actionable insights and support data-driven decisions. I communicate findings clearly to diverse audiences, thrive in collaborative environments, and am passionate about mentoring and continuous learning.

EDUCATION

African Institute for Mathematical Sciences (AIMS) – Senegal **Oct 2025 – Present**
Master of Science. Data Science (Cooperative Programme)

Jaramogi Oginga Odinga University of Science and Technology – Kenya **Aug 2019 – Mar 2024**
Bachelor of Science. Actuarial Science with IT

PROJECT EXPERIENCE

Air Quality Index Prediction – Comparative Machine Learning Study **2026**
AIMS MSc Project | Dakar, Senegal

- Compared ML algorithms (Random Forest, XGBoost, SVR, Neural Networks) to predict AQI in Dakar, Senegal.
- Built end-to-end pipeline: data cleaning, feature engineering, hyperparameter tuning, and model evaluation.

Smart City Air Quality Forecasting – Sensor Placement Optimization **2026**
AIMS MSc Project | Nairobi, Kenya

- Designed an AQI forecasting system combining time-series models with submodular greedy and convex optimization.
- Determined cost-effective sensor placement locations to maximize air quality monitoring coverage from 24 sensor to 12.

West African Agricultural Market Intelligence **2025**
AIMS Group Project (WAAMC) | Nigeria, Ghana, Senegal, Burkina Faso

- Analysed crop price trends, trading volumes, and market dynamics across multiple countries.
- Generated data-driven insights supporting farmer decision-making and regional food security.
- Collaborated across cross-country teams to deliver policy-relevant outputs.

PROFESSIONAL EXPERIENCE

Kenya Marine and Fisheries Research Institute (KMFRI) – Kenya

May – Aug 2023

Data Analyst Intern

- Analysed socioeconomic and fisheries data using R programming, SPSS, STATA, and Excel
- Designed and conducted surveys for primary data collection on lake region livelihoods
- Cleaned, validated, and integrated multi-source datasets for analysis.
- Prepared structured datasets for Power BI and Tableau dashboards and reporting.
- Collaborated with researchers and stakeholders to inform fisheries policy.

TECHNICAL SKILLS

- **Languages & Querying:** Python, R, SQL/PostgreSQL, Java, C/C++, LaTeX.
- **Machine Learning & AI:** Scikit-learn, PyTorch, TensorFlow, Random Forest, SVM, NLP, Computer Vision, Generative AI.
- **Data Tools:** Power BI (dataset preparation & dashboards), Tableau, Excel (Power Query, Pivot Tables, Advanced Formulas), SPSS, STATA, Kobo Toolbox, Google sheet.
- **Other:** Git/GitHub, Linux, Web Scraping, Data Pipelines, Database Systems, Numerical Optimization.

SOFT SKILLS

- Effective communicator and award-winning presenter, team player and group leader, complex problem-solver, adaptable and continuous learner, cross-cultural collaboration (Kenya, Senegal, West Africa)

AWARDS AND SCHOLARSHIPS

- Mastercard Foundation Scholarship - MSc. Data Science, AIMS Senegal (2025 – Present).

HIGHLIGHT OF GRADUATE COURSEWORK

Programming with Python, Linear Algebra and Applications, Probability and Statistics, Mathematical Problem Solving, Numerical Optimization, Data Collection, Data Preprocessing, Database for Data Sciences, Statistical Mechanics, Machine Learning, High Dimensional Data Analysis, Data Mining and Big Data Analytics, Deep Learning and Neural Networks & Big Data, Graph Theory and Applications, Computer Vision, Generative AI, Mathematical Methods for Climate Science, and Climate Dynamics.

LANGUAGES

- English – Fluent | Swahili – Native | French – Beginner

REFERENCES

Available on request.